

PSU ARCHITECTURE

Contents

Introduction 2

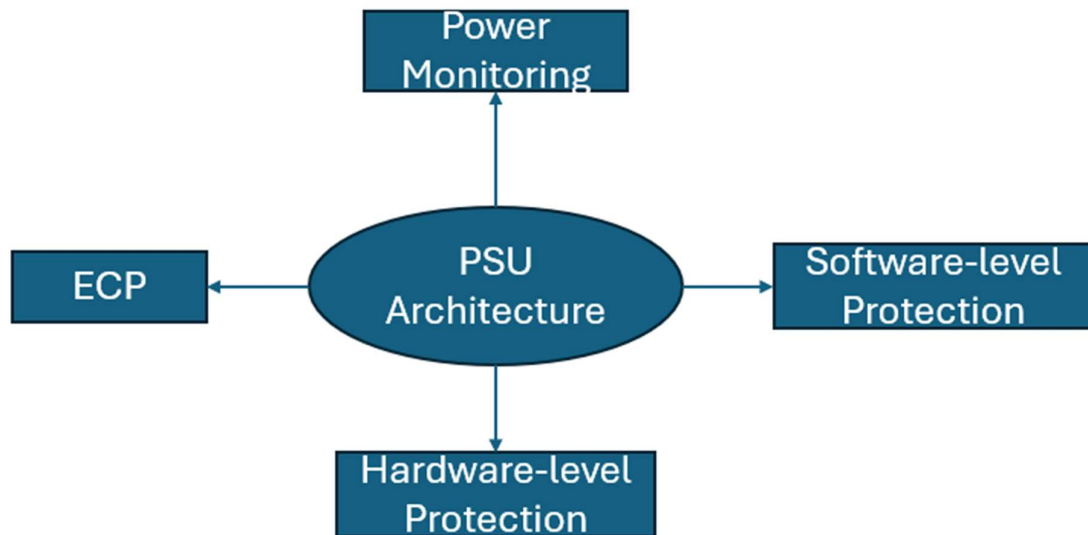
PSU Architecture Main Design 3

PSU Architecture Trade-offs 4

Introduction

This PSU architecture is designed to make the PSU more efficient and reliable. This PSU architecture was created as the PSU is the main component for electrical safety within a device. This means that if it isn't reliable enough then the device can take damage or stop functioning entirely. This PSU architecture does not change the main processes that current PSUs use. For example, it does not fundamentally change the input stage, PFC, switching stage or output stage. This PSU architecture focuses on communication between the PSU and rest of the system. It also focuses on improving electrical safety of the device.

This PSU architecture cannot promise perfect efficiency or reliability as it is impossible to achieve in any system. However, this PSU architecture promises to try provide a more efficient and reliable PSU compared to today's current models. This PSU architecture was made to be used on most devices that are used today. The technical specifications in this architecture are made to help engineers develop this PSU architecture more easily. Make sure to adapt this PSU architecture to real-world standard and resources.



The spider diagram above shows the 4 main concepts involved within this PSU architecture. ECP stands for electric control protocol which will be used to enforce electrical safety. You can find more information in the sections below.

PSU Architecture Main Design

- It is important that the PSU be able to protect the device's components from being fired by high voltages.
- I will first be discussing how ECP will work.
- First, the motherboard calculates how much voltage each component needs to meet the power requirements.
- Second, the motherboard uses current-sense resistors which are placed in each component's power path to find each component's power requirements.
- Third, the motherboard measures the tiny voltage drop and uses Ohm's law to calculate providing an instantaneous reading.
- Fourth, the motherboard sums up the total voltage needed adds a 5% safety buffer and sends the request to the PSU.
- Fifth, the PSU adapts and lowers or increases the voltage as needed to ensure that the power requirements are met exactly.
- There will also be a maximum electricity limit that the PSU can delivery to prevent overcurrent or overflow.
- This system uses 4 tiers of maximum and minimum electricity limitations.
- Tier 1 has a minimum of 50W and maximum of 200W, Tier 2 has a minimum of 200W and maximum of 400W, Tier 3 has a minimum of 400W and maximum of 1200W and Tier 4 has a minimum of 1000W and 2000W.
- The user can set a limit within these ranges from the BIOS and from other firmware options for more control.
- This electricity maximum limit will be enforced at both the hardware and software level.
- For the software-level, ECP will be used to enforce the limit as the motherboard will be monitoring the amount of electricity the PSU is delivering.
- For the hardware-level, a current limiting circuit will be used to enforce the limit even if the software-level enforcement were to fail.
- If the electricity limit were to be reached above, the PSU will immediately reduce the voltage to the limit itself or below it.
- It will not cut off power as that could make users lose data or cause damage.
- This improves electrical safety as it ensures that the PSU cannot delivery electricity that would damage the device.
- I will now be discussing power monitoring
- First, the PSU uses current sense resistors to measure how much electricity it is currently delivering.
- Second, this reading is sent to the motherboard so the motherboard can use it for adapting to workload.
- Third, the motherboard uses these readings to ensure that it doesn't repeat requests for the same power delivery.
- This improves efficiency as it prevents confusion in power delivery within the PSU as the motherboard knows exactly what it is dealing with.

PSU Architecture Trade-offs

- **This PSU architecture has trade-offs just like any other system I have created.**
- **Engineers and developers may also face further implementation challenges as each system is different.**
- **I have identified 5 main trade-offs.**
- **The first trade-off is time-consumption.**
- **This trade-off exists because it will take years to implement these hardware changes to the PSU.**
- **This means that the PSU will not be a short-term project and will require consistent resources and effort.**
- **This trade-off can be managed by using structured development processes, working in teams and using existing systems and methods.**
- **The second trade-off is cost.**
- **This trade-off exists because it will mostly likely cost a lot to implement the hardware changes.**
- **This is because implementing resistors and circuits cost physical materials and resources to create.**
- **This trade-off can be managed by using resistors and circuits only where needed, allocating a budget for this project and accepting that this will have its costs.**
- **The third trade-off is coordination.**
- **This trade-off exists because this PSU architecture will require hardware engineers, electrical engineers and OS developers to work together.**
- **This means that there must be a long-term commitment from multiple parties to create this PSU.**
- **This trade-off can be managed by using existing business agreements, clear roles and responsibilities and central oversight to prevent conflicts.**
- **The fourth trade-off is complexity.**
- **This trade-off exists because this PSU architecture can be difficult and confusing to understand and implement.**
- **This is because it requires changes to multiple components like firmware, the motherboard and the PSU itself.**
- **This trade-off can be managed by understanding the architecture, using structured project management processes and consistently testing concepts.**
- **The fifth trade-off is performance.**
- **This trade-off exists because this PSU architecture has limits it can not push all the way to its max form.**
- **This means that power users will not be able to push performance to his limits to the point that it can cause damage.**
- **This trade-off can be managed by allowing users to control voltage within safe limitations and warning them that they may will not be able to push it beyond limits.**